

GLASGOW COLLEGE UESTC

Main Exam Paper

Power Electronics (UESTC3022)

Date: June 15th 2023

Time: 14:30-16:30PM

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

For Multiple Choice Questions, use the dedicated answer sheet provided.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

DATA/FORMULAE SHEET IS PROVIDED AT THE FRONT OF PAPER

Power Electronics 2 Formulae List

For an inductor: $v_L(t) = L \frac{di_L(t)}{dt}$, $i_L(t) = \frac{1}{L} \int v_L(t) dt$, $E_L(t) = \frac{1}{2} L [i_L(t)]^2$

For a capacitor: $v_C(t) = \frac{1}{C} \int i_C(t) dt$, $i_C(t) = C \frac{dv_C(t)}{dt}$, $E_C(t) = \frac{1}{2} C [v_C(t)]^2$

Average value: $V_{ave} = \frac{1}{T} \int_0^T v(t) dt$

RMS value: $V_{rms} = \sqrt{\frac{1}{T} \int_0^T v(t)^2 dt}$

Duty cycle: $D \equiv \phi = \frac{T_{on}}{T_{on} + T_{off}} = \frac{T_{on}}{T_{period}}$

Amplitude modulation index of SPWM:

$$m_a = \frac{\hat{V}_{control}}{\hat{V}_{carrier}}$$

Frequency modulation index of SPWM:

$$m_f = \frac{f_{carrier}}{f_{control}}$$

Instantaneous power:

$$p(t) = \frac{v^2(t)}{R} = i^2(t)R = v(t) \times i(t) = \text{force} \times \text{velocity} = \text{torque} \times \text{angular velocity}$$

Average power:

$$P = \int_0^T v(t)i(t)dt = \frac{V_{rms}^2}{R} = I_{rms}^2 R$$

reverse recovery power loss in a diode:

$$P_{RR} = Q_{RR} \times V_{R(blocking)} \times f_{switching}$$

MOSFET switching loss:

$$P_{switching} = \frac{f_{switching} V_{DS(off)}}{2} (T_{on} I_{on} + T_{off} I_{off})$$

Thermal circuit:

$$\Delta T = P_{device} \times \Sigma \theta_{XY} \quad \text{or} \quad \Delta T(^{\circ}\text{C}) = \frac{\text{power} \times \text{time}}{\text{thermal capacitance}}$$

Junction temperature of device:

$$T_{junction} = T_{sink} + P_{device} (\theta_{junction-case} + \theta_{case-sink})$$

Calculus:

$$\cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\int \cos(a\theta) d\theta = \frac{1}{a} \sin \theta \quad \int \cos(a\theta) d\theta = -\frac{1}{a} \sin \theta$$

Q1. Answer all questions in this section:

(a) The sawtooth voltage function $v(t)$ is shown in Figure Q1(a), whose period is 10 ms and peak value is 200 V.

- i. Please draw $v(t)$ full-wave rectified waveform, i.e., $|v(t)|$ waveform. [1]
- ii. Calculate the average voltage of $|v(t)|$. [1]
- iii. Calculate the RMS voltage of $|v(t)|$. [2]
- iv. Calculate the form factor of $|v(t)|$. [1]

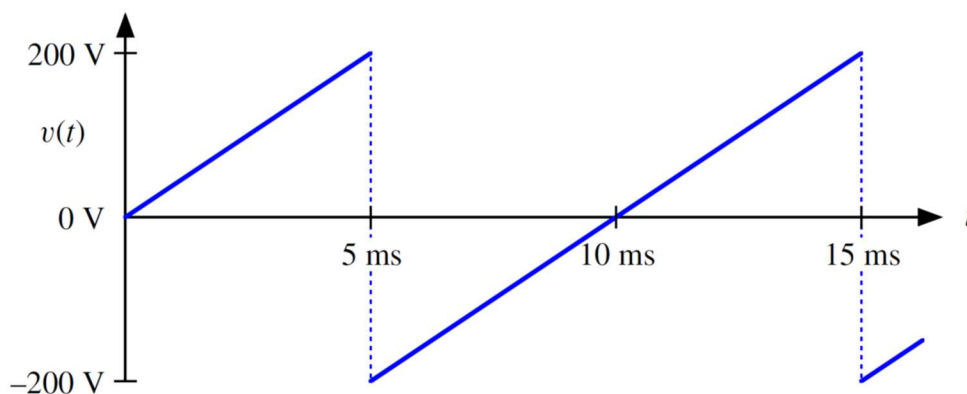


Figure Q1(a)

(b) A power diode is used in a switched-mode power supply (SMPS) circuit which is operating at 20 kHz. The reverse voltage across the diode during the pulse-width modulation (PWM) “Off” period is -50 V. Using the additional data in Table Q1, calculate the total power dissipated in the diode. [5]

Table Q1

Parameters	Values
Forward current (I_F)	5 A
Forward voltage drop @ $I_F = 5$ A (V_F)	1.1 V
SMPS duty cycle (ϕ)	50.0%
Reverse recovery charge (Q_{RR})	2.5 μ C

(c)

- i. Draw and label a circuit to show how two silicon controlled rectifiers (SCRs) can be used to control a resistive load from an AC supply, allowing current to flow in both directions through the load. [2]

- ii. Draw and label sketches of the current through the load to show how phase-angle control of SCRs is used to provide an average power (i) less than 50% of its maximum value and (ii) above 50% of its maximum value. [3]
- (d)
- i. A bridge rectifier is required to supply an average current of 200 mA with a ripple not to exceed 0.5 V. What size of smoothing capacitor is needed? The AC supply is at 50 Hz and the capacitor supplies the current for only 60% of the time.
- ii. A transistor dissipates 2 W and has a thermal resistance of $5^{\circ}\text{C}/\text{W}$ between the junction and case. Calculate the junction temperature when the ambient temperature is 50°C with a heat sink of thermal resistance (including the contact) of $50^{\circ}\text{C}/\text{W}$.
- (e) Consider the circuit in Figure Q1-e. Assume the power diodes D_1 , D_2 , D_3 , and D_4 are ideal diodes, i.e., their forward voltage drop = 0 V. Determine the following when a $V_{AC} = 25 \sin(100\pi t)$ voltage is applied to the primary winding of the transformer:
- i. What is the peak output voltage for the bridge rectifier? [2]
- ii. Draw the voltage waveform across R_L [2]
- iii. What is the minimum peak inverse voltage (PIV) for each diode? [1]

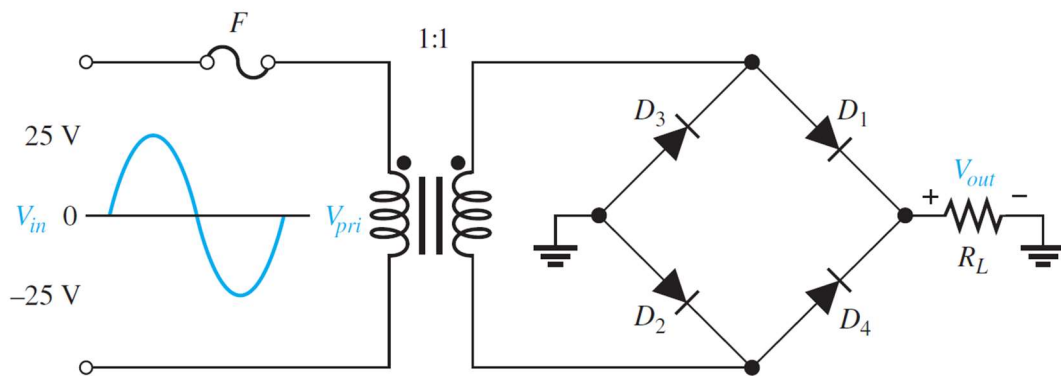


Figure Q1-e

- Q2. Fig.Q2 shows a MOSFET acts as a low-side switch for a load whose power is controlled by pulse-width modulation (PWM). The parameters of the system are given in table Q2. Assume that components are ideal and that the current increases linearly while the MOSFET is on.

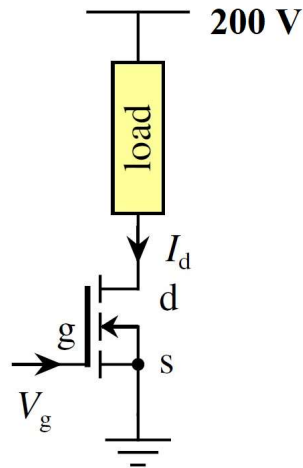


Fig.Q2

Table Q2 Parameters for a load controlled by a MOSFET using PWM

Parameters	Values
I_{ON}	16 A
I_{OFF}	24 A
PWM Duty cycle	50%
PWM Frequency	50 KHz
Supply voltage	200 V DC
$R_{DS(on)}$	50 m Ω
T_{ON}	25 nS
T_{OFF}	30 nS

- Draw the current waveform. Clearly indicate the current and time values. [4]
- Calculate the average and RMS value of the current. [7]
- Determine the conduction loss in the MOSFET. [4]

- (d) Determine the switching loss in the MOSFET? [4]
- (e) Determine the total power dissipated in the MOSFET. [2]
- (f) The power loss is found to be excessive so the designer suggests halving the switching frequency, keeping other parameters the same. Will this be effective? [4]

Q3 A converter circuit is shown in Figure Q3 for adapting a 20 V/50 kAH battery supply to a laptop which needs 10 V. Consider the converter is operating in continuous mode at the boundary condition, with the input voltage V_{IN} being 20 V, the output load resistance R_L being 10 Ω and the switching frequency being 50 kHz.

- (a) What is meant by “continuous mode” at the boundary condition? [2]
- (b) What type of converter is it? Derive the expression of the output voltage as a function of the input voltage and calculate the required switching duty cycle. [8]
- (c) Assuming the value of the capacitance C is sufficiently large, determine the value of inductance L for the converter operating in continuous mode at the boundary condition. [4]
- (d) Sketch the voltage and current waveforms of the inductor, the current of the diode, and the current waveform of the capacitor. Axes should be clearly labeled with right scales. State any assumptions you make. [8]
- (e) Name another converter that can be used by the engineer to power the laptop, give the expression for the output voltage as a function of the input voltage, and calculate the required switching duty cycle. [3]

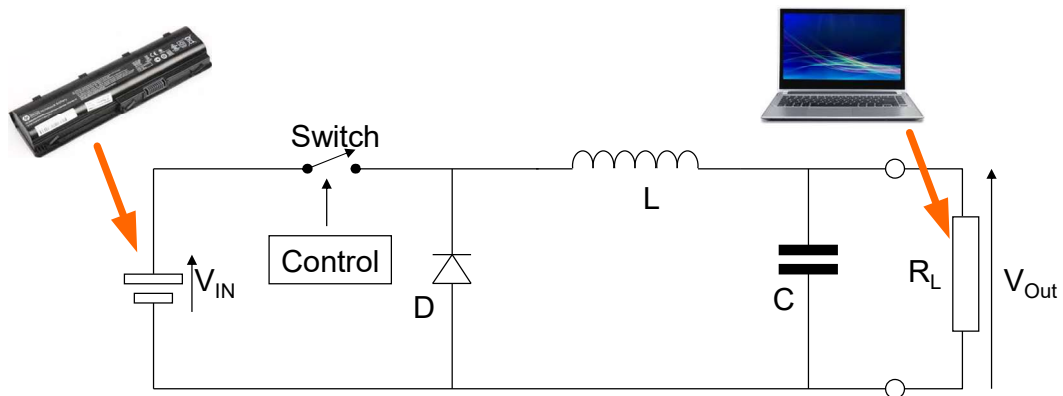


Figure Q3

- Q4 Figure Q4 shows a three-phase PWM inverter with its bipolar PWM drive circuit, where the carrier frequency f_c is much higher than the control signal frequency f , the control signals $v_a = m_a \sin(2\pi ft)$, $v_b = m_a \sin(2\pi ft + 2\pi/3)$ and $v_c = m_a \sin(2\pi ft + 4\pi/3)$, and peak values of the triangle carrier signal are 1 and -1.
- If $0 < m_a \leq 1$, determine output three-phase line-to-line voltages v_{AB} , v_{BC} and v_{CA} , neglecting their high frequency harmonics components. [9]
 - What is the 'square-wave' operation mode of the inverter? Indicating its benefits and disadvantages. [4]
 - What are the advantages of a three-phase inverter over a single-phase inverter? [4]
 - How to change the phase sequence of the three-phase inverter? [4]
 - What is the 'shoot-through' of the inverter? How to take precaution measures to avoid 'shoot-through'? [4]

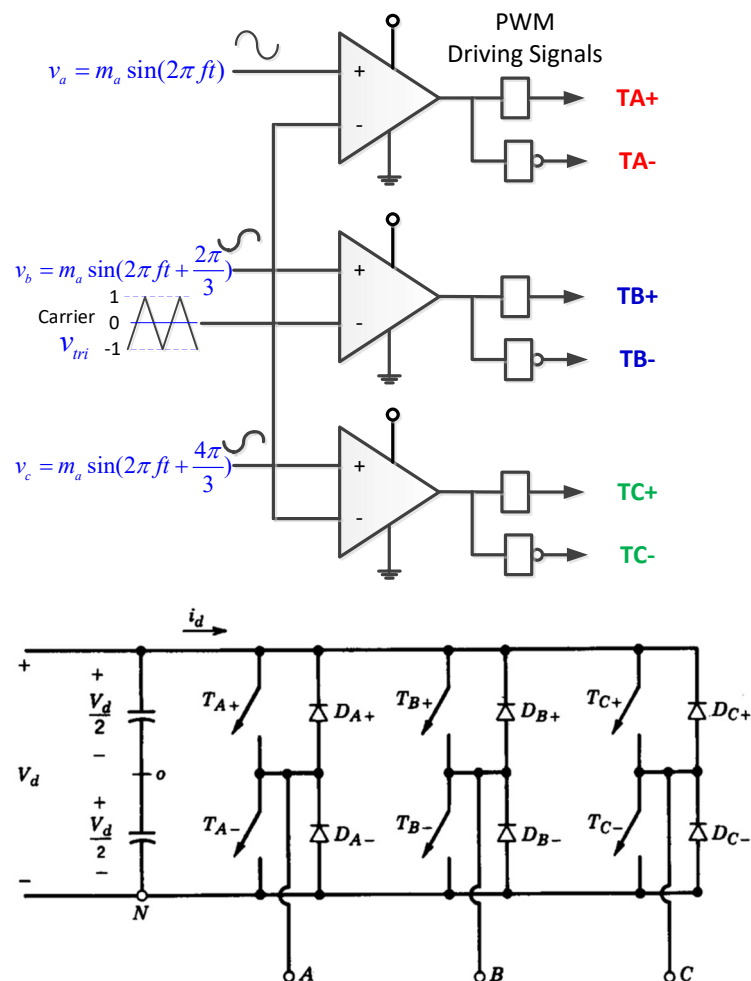


Figure Q4